AI Communication Systems Study Kit

College Project Report

Project Repository:
github.com/madhumithasudhakar969-cyber/AI_Communication_Systems_Study_Kit

Student Name: _____

Department / Class: _____

College: _____

Submitted to: _____

1. Abstract

This project is a beginner-level Python notebook that introduces communication-system signal visualization. It prints a welcome message and uses NumPy to create signals and Matplotlib to plot two modulation examples: Amplitude Modulation (AM) and Frequency Modulation (FM). The notebook can help students observe how a carrier signal behaves when it is modulated by a message signal.

2. Objectives

• Introduce a simple Python notebook workflow.
• Generate time values and sinusoidal signals using NumPy.
• Visualize an Amplitude Modulation (AM) signal.
• Visualize a Frequency Modulation (FM) signal.
• Support basic understanding of communication-system concepts through graphs.

3. Tools and Technologies

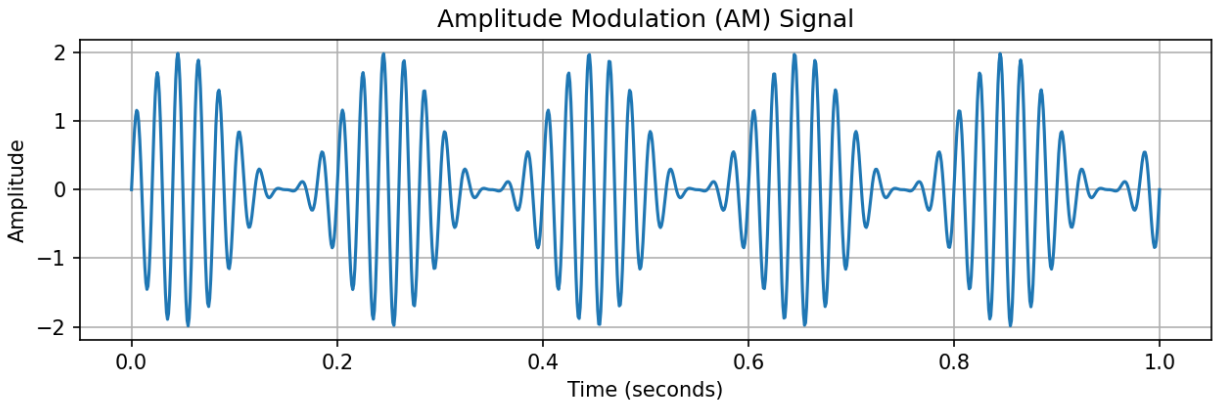
• Python 3
• Jupyter Notebook / Google Colab
• NumPy — numerical arrays and signal calculations
• Matplotlib — plotting and visualizing signals
• GitHub — storing and sharing the notebook

4. Project Workflow

1. Open the notebook from the GitHub repository.
2. Run the introductory cell to display the project welcome message.
3. Import NumPy and Matplotlib.
4. Create a time array and message/carrier signals.
5. Calculate the AM signal and display its graph.
6. Calculate the FM signal and display its graph.

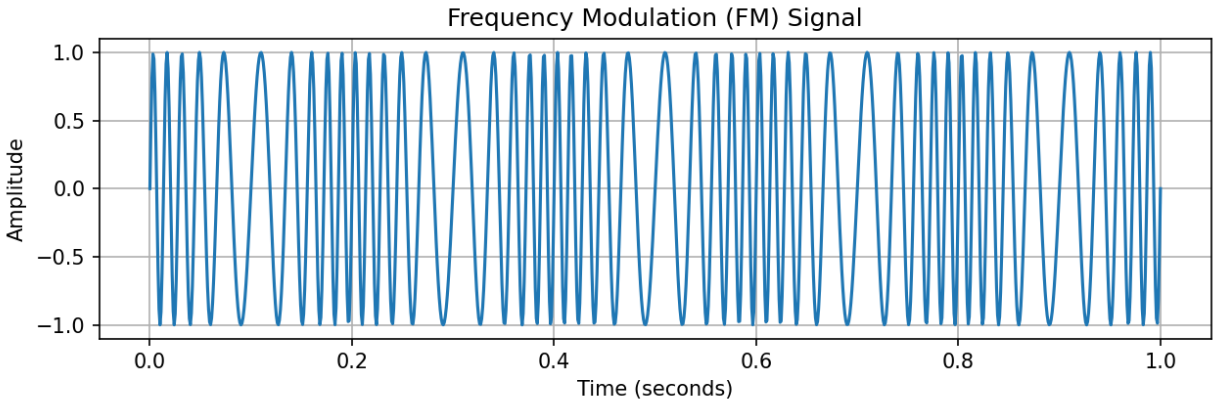
5. Amplitude Modulation (AM)

Amplitude Modulation changes the amplitude of a high-frequency carrier according to the message signal. In the uploaded notebook, the message frequency is 5 Hz and the carrier frequency is 50 Hz. The example signal is calculated using: **AM = (1 + message) × carrier**.



6. Frequency Modulation (FM)

Frequency Modulation changes the instantaneous frequency of the carrier according to the message signal. In the notebook, the message frequency is 5 Hz, the carrier frequency is 50 Hz, and the modulation index beta is 5. The example signal is calculated using: **FM = sin(2πfc t + β sin(2πfm t))**.



7. Expected Output

When the notebook cells run successfully, the first cell prints a welcome message. The AM and FM cells each display a graph of the generated signal. These graphs are the main visual outputs of the current notebook.

8. Conclusion

The project demonstrates a basic simulation of AM and FM signals using Python. It connects programming with a core topic in communication systems and gives students a visual starting point for understanding modulation. Future improvements could include a user interface, explanations beside each graph, and interactive controls for changing frequencies and modulation parameters.

9. Repository

GitHub repository: https://github.com/madhumithasudhakar969-cyber/AI_Communication_Systems_Study_Kit

Submission note: This report is based on the notebook currently visible in the public repository. Please add your name, class, college, and teacher details, and confirm that the notebook runs correctly before submitting.